INTEL

UNIVERSITY VLSI BOOTCAMP SERIES 2026

SMART CAR PARK SYSTEM

Objectives

- Work in a team and collaborate effectively to solve design problems.
- Understand and design a document of HAS (high-level specifications) of a system-on-chip.
- Communicate, explain, and discuss the proposed designs.

1 Background

The increasing number of vehicles in urban and residential areas has created major challenges in parking management. Conventional parking systems often rely on manual monitoring or basic entry and exit control, which may lead to traffic congestion, inefficient space utilization, and difficulty for drivers in locating available parking spaces. Recent developments in smart parking systems have demonstrated that intelligent monitoring, real-time parking availability, and convenient digitalized services can improve parking efficiency and enhance user experience [1].

2 High Level Overview

This document describes the architecture of a SoC (System-on-Chip) for a **Smart Car Park System**. The Smart Car Park System is intended to improve and optimize vehicle parking management in public, commercial, or residential parking areas. The system is designed to **sense car entry and exit, control barrier gates automatically, monitor parking occupancy and provide parking information in real time**.

[*Disclaimer: the final product is a high-level specification document; no coding is required*]

3 Design Requirements

The SoC for the Smart Car Park System should support at least (but not limited to) 1 counter, 2 input units and 2 output units. The SoC should also have built-in register files and memory for storing system data, parking records, and operational logs. Additional marks will be given to teams who successfully implement relevant smart or advanced features.

3.1 Basic functions of Smart Car Park System

3.1.1 Counter

- Counters can be used to perform basic functions, such as counting the number of occupied parking spaces and remaining parking spaces based on the input sensor.

3.1.2 Input Units

- Camera.
- Emergency Button.

3.1.3 Output Units

- Barrier motor control.
- Entry/exit kiosk display.

3.2 Additional Advanced Feature

The system should include **at least one** advanced feature. Open for designers to add any features out of the list. Few examples are as below:

- Cashless payment.
- Lane clear sensor (*to detect that no vehicle is in front of the barrier before the barrier opens*).
- Mobile application for parking reservations.
- Car break-in detection using image recognition.
- **Not limited to the above examples, encouraged to add based on group discussion.**



Figure 1: Smart Car Park Illustration [2]

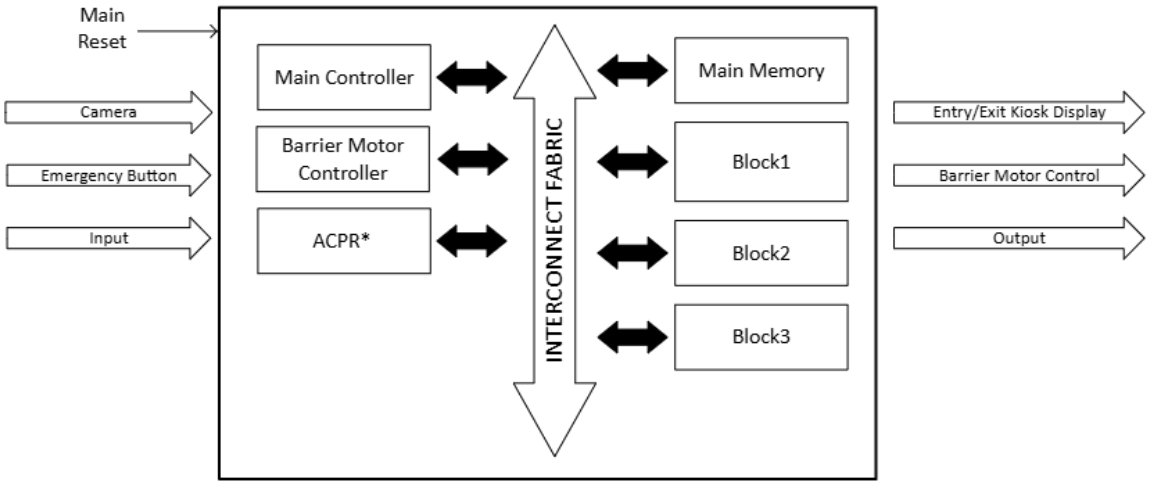
4 Challenge

Complete all 7 challenge subsections (4.1 – 4.7). Please ensure that your design choices are consistent with the architectural details presented and supported by adequate justification.

4.1 Block Diagram

The block diagram below is an example with some internal blocks already defined. Define all the possible I/O's, insert the internal IP design blocks and show how each of them is related to the inputs, internal blocks, and outputs. [10 marks]

[*Note: You may add more boxes and I/O as necessary.*]



**ACPR – Automatic Car Plate Recognition*

4.2 Subsystem / IP Design Blocks

Identify the necessary IP designs and describe their functions. Several example subsystems are provided below. Complete the rest of the subsystems and their description below. [10 marks]

Subsystem/Module	Description
Main Controller	Controls and manages vehicle entry and exit operations in the Smart Car Park, SoC.
Barrier Motor Controller*	Controls barrier movement, including lifting and lowering operations.
ACPR*	Recognizes vehicle license plate numbers using the Automatic Car Plate Recognition system.
LCD/Display Controller*	Display control module for presenting parking information at entry and exit kiosks, such as space availability, timestamps, and parking fees.

**Can treat as black box, do not need implementation details.*

[*Useful hints: Q4.1 and Q4.2 must be defined first because it will affect all the subsequent answers.*]

4.3 Interface Signals List

Define the I/O signal interfaces of the SoC. Two signal interfaces are listed below. Complete by adding more signal interfaces and descriptions below. [10 marks]

Signal	Direction	Description
Camera	Input	Captures vehicle license plate images at the entrance and exit points.
Emergency Button	Input	To request assistance (*Eg: Barrier stuck, card not scannable, payment failure, etc.*)

[*Useful hints: Outputs of sensors are inputs to the SoC*]

4.4 Storages

4.4.1 Memory

List **3 or more memory data** to be stored in non-volatile memory which are applicable in this SoC.

[3 marks]

Data	Description
Eg: Car plate number	Recognized car plate number for each entry
Eg: Entry Timestamp	Entry timestamp for each vehicle

[*Useful hints: Non-Volatile refers to data that are retained even if the system is powered off*]

4.4.2 Control Registers List

Define all the control registers to be used in the SoC. Examples are given below. Complete it by adding more registers and their description. [7 marks]

[*Note: Flexibility is allowed in defining the CR type, bit width, and field allocation, provided the overall design is consistent and practical.*]

[*Note: Reset value means the initialization value to set for each field after CR is out of reset.*]

[*Note: You can create any number of CR (R1, R2, R3…) according to your design needs, following the Template below.]*

Example:

R0: GENERAL_IN <Bit Width = 16>

Field Name	Bit	Type	Reset Value	Description
ENTRY_BUTTON	0	RW	1'b1	Turn on or off the assistance button for car park entry kiosk. 1’b1 to turn on, 1’b0 to turn off.
EXIT_BUTTON	1	RW	1'b1	Turn on or off the assistance button for car park exit kiosk. 1’b1 to turn on, 1’b0 to turn off.
RSVD	15:2	RO	12'b0	Reserved. For future extension.

[Note: RO: read_only WO: write_only RW: read_write]

Template: <Insert CR Name>

Field Name	Bit	Type	Reset Value	Description

[*Useful hints: Think of RO, WO and R/W from the perspective of the main controller/core*]

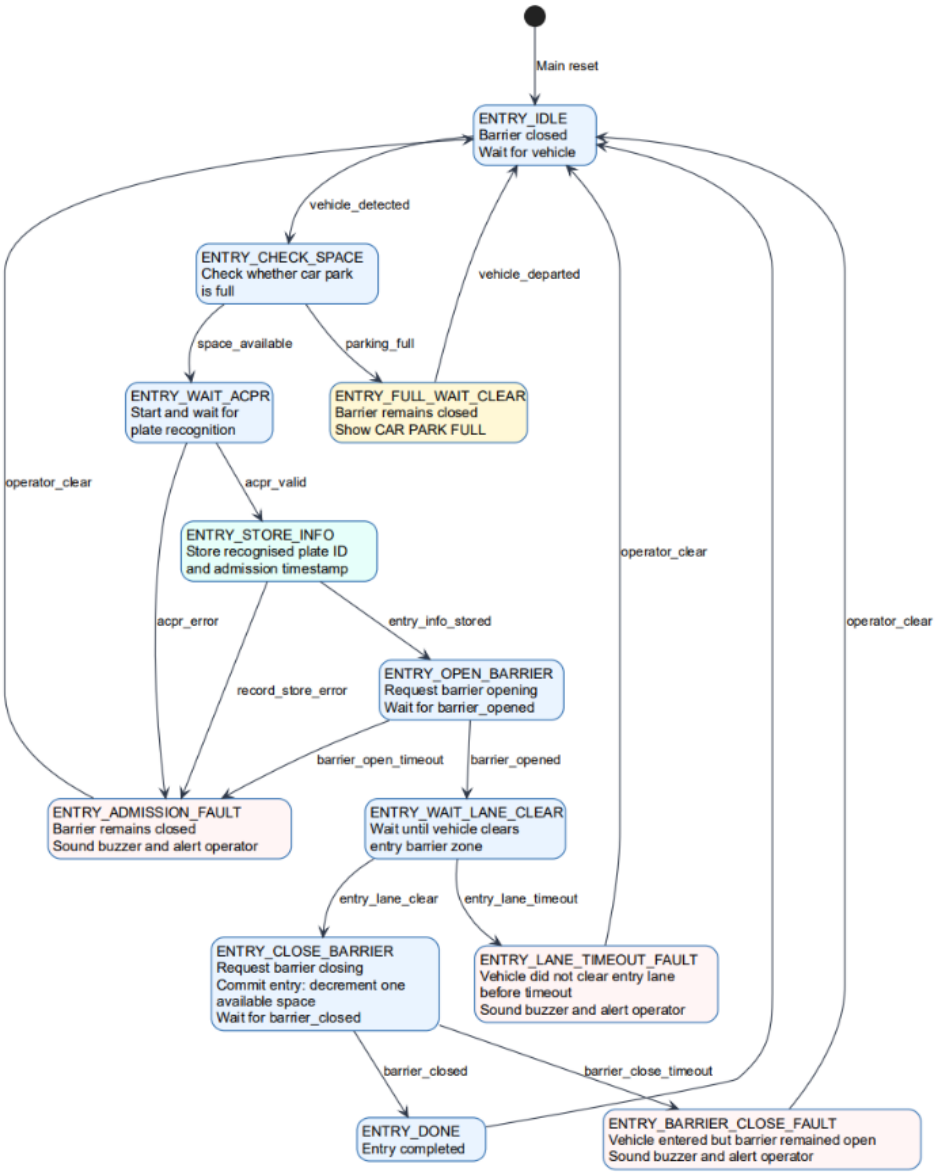
4.5 SoC FSM Implementation

The Main Controller will manage the vehicle entry and exit operations. Based on the architectural definitions provided in Sections 4.1 to 4.4, you are required to develop the corresponding exit finite state machines (FSMs).

a. Draw an FSM for car park exit operation. (*The expected car park entry FSM is shown below. You may use it as a template to extend your design for car park exit FSM*) [10 marks]

b. Document in a table and briefly describe each state. [5 marks]

Below shows an FSM example of a car park entry FSM:



State	Descriptions
ENTRY_IDLE	Barriers are closed. Wait for the vehicle.
ENTRY_CHECK_SPACE	Check whether the car park is full.
ENTRY_WAIT_ACPR	Start and wait for car plate recognition.
ENTRY_STORE_INFO	Store recognized plate ID and admission timestamp.
ENTRY_OPEN_BARRIER	Request for barrier opening. Wait for barrier_opened.
ENTRY_WAIT_LANE_CLEAR	Wait until vehicle clears entry barrier zone.
ENTRY_CLOSE_BARRIER	Request barrier closing. Commit entry: decrement of one available space. Wait for barrier_closed.

4.6 System Validation Test Plan

Provide at least **4 directed and 3 random test** conditions that should be added to the test plan. Example as below.

Example: Power on the SoC, assert reset and verify that all IPs and registers are correctly initialized before the first car entry or exit activity.

Directed Test

1.

2.

3.

4.

[4 marks]

Random Test

1.

2.

3.

[6 marks]

4.7 DFx Implementation

1. Based on the design,
    a. Provide 2 examples of potential hardware issues that could potentially be found on the Smart Car Park system. [2 marks]
    b. How can Design for Excellence (DFx) features help to detect the issues mentioned in (a)? (In terms of debug hook) [2 marks]

2. In your opinion, from the DFx perspective, what are the possible ways that can help improve the speed of testing of a design?

    [1 mark]

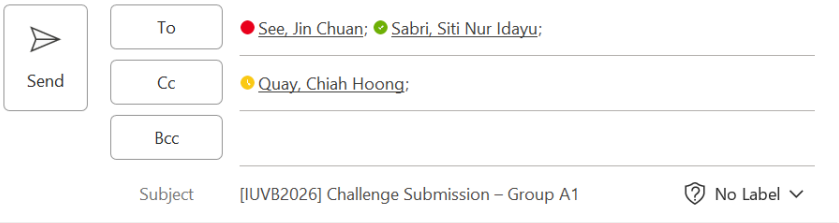
5 Presentation Requirements [Please read carefully]

Your answers to the challenge should follow the criteria below:

- Answers should be documented in presentation form using **Microsoft PowerPoint;** the slides styles are up to your team.
- Other softwares/applications (eg: Microsoft Word, Vision, etc.) are allowed to be used for any complicated/advanced illustration/figures during presentation. If you are doing so, kindly take a snapshot of your illustration and submit it together with your slides. You may also attach it at the end of your pptx as appendix.
- **Kindly rename your PowerPoint slides using this format**,
    - Challenge_Group_<group_name>.pptx (eg: Challenge_Group_A1.pptx).



- **Kindly use this as your email subject/title:**
    - [IUVB2026] Challenge Submission - <replace_with_group_number>
        - Eg: [IUVB2026] Challenge Submission – Group A1



- Submit latest by email before **1.00 pm on 25 October 2026**, any late submission will be penalized with 5 marks reduction of your total scores.
    - To: jin.chuan.see@intel.com; siti.nur.idayu.sabri@intel.com
    - Cc: chiah.hoong.quay@intel.com
- Each group will have 15 minutes of presentation time; 5 minutes of Q&A and presentation order is random and up to the judges' preferences!

6 References

[1] S. S. Channamallu, S. Kermanshachi, J. M. Rosenberger, and A. Pamidimukkala, 2026. Smart parking systems: A comprehensive review of digitalization of parking services. https://www.sciencedirect.com/science/article/pii/S277315372500043X#bib91

[2] Adapted from https://rb.gy/kv4rsn

Appendix

Challenge Session Assessment Criteria:

Design Aspects	Points
Drawing of I/O and Subsystems	10
Subsystems Functional Description	10
I/O Description	10
Storages (Memory & CR)	10
Finite State Machine	15
VALIDATION	
Test Plan Writing	10
DFX	
DFx Implementation	5
INNOVATION	
Novel Ideas • Design or corner cases ideas.	20
Soft skills Assessment (Judges) • Presentation skills. • Time management. • Team player / team spirit. • Involvement in discussion / response during Q&A.	10
Total	**100**

[Note: Design choice justification should be taken into consideration when developing all design aspects. Impress the judges by telling the story behind your design choices!]

Challenges in life are doorways for improvements.